

Airframe Technology Master

Fatigue and Damage Tolerance of metallic and composite structures

Unit 1

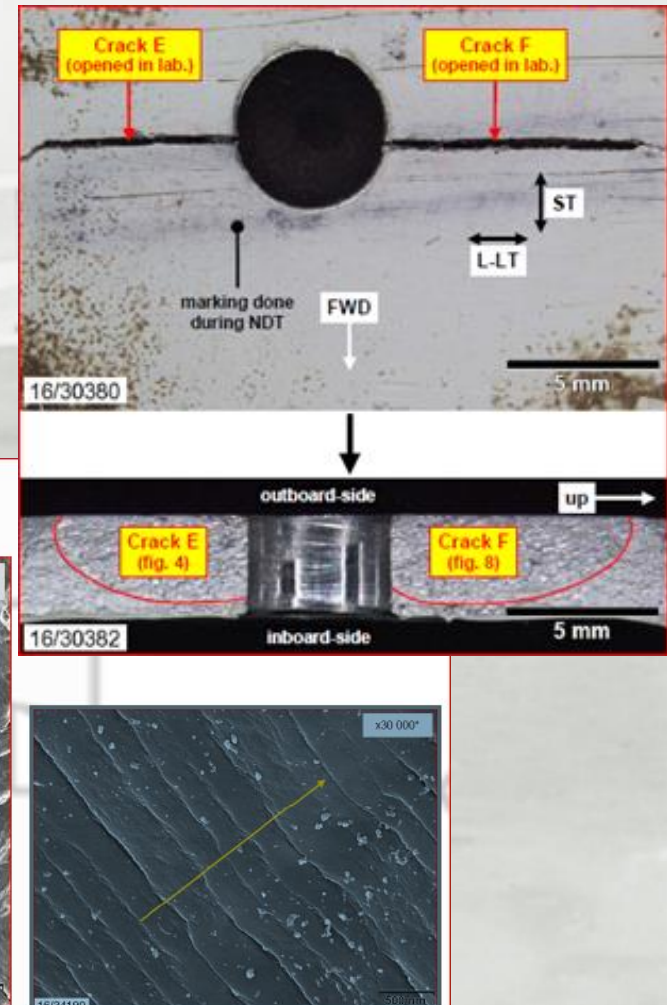
José I. Armijo

Efraín Mirón

2nd June 2025

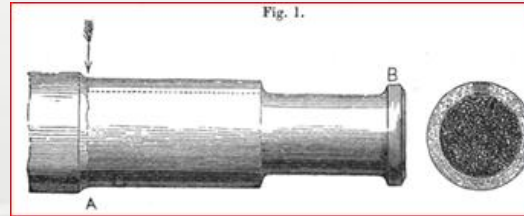
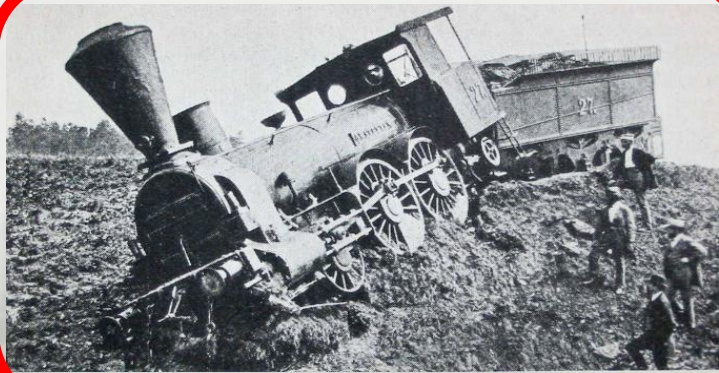
Unit 1: Introduction. History. Concepts. Definitions. Standards.

- History of structural fatigue (20 min)
- Introduction (80 min)
 - Definition of usual terms
 - Applicable requirements and standards: FAR/JAR
 - Standards applicable to military aircraft
 - Primary basic concepts of fatigue (allowables)



Fatigue fracture – crack growth direction and striations

History of structural Fatigue



FIRST FATIGUE STUDIES

First reported in-service accident due to a crack in the axle of a railway wheel (France, May 1842)



August Wöhler, manager of “Imperial Railways” of Germany during 1874 to 1889 is considered the first person to undertake a systematic investigation of the fatigue phenomenon.

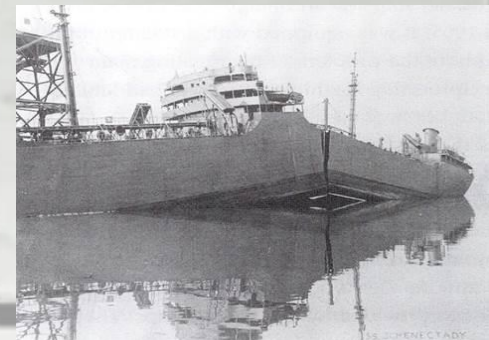
VI. *The Phenomena of Rupture and Flow in Solids.*
By A. A. GRIFFITH, M. Eng. (of the Royal Aircraft Establishment).
Communicated by G. I. TAYLOR, F.R.S.

Received February 11,—Read February 26, 1920.



First publication about fracture mechanics was performed by Griffith in 1920.

During the second World War, cracks were detected on more than 25% of the welded Liberty ships built in the United States.



History of structural Fatigue



Static Strength

Resistance against very high loads (Design Static Loads) that have low probability of occurrence.

Fatigue Strength

Resistance against repeated loads. Loads much lower than the Design Static Loads may produce structural failure if repeated sufficiently often.

“I am constructing my machine to sustain about 5 times my weight and I am testing every piece. I think there is no possible chance of its breaking while in the air”, Wilbur Wright (1900)

...Yet...During test-phase the propeller shaft failed due to ‘fatigue’

The first flight of the Wright brothers was delayed due to cracks on the propeller axle.

History of structural Fatigue

Background

The Stress-Fatigue and Damage Tolerance (F&DT) discipline is a **key element in the Design, Certification and Continued Airworthiness of aircraft**, and it is one of the **core** competences of any aerospace industry since it affects safety and economic aspects throughout the whole life cycle of the aircraft. To address the structural integrity of the Airbus Defence and Space aircraft, as they age, is a paramount task.

Aircraft **aging** is one of the main concerns of industry and authorities; the regulations are **under permanent scrutiny and revision** in order to maintain safety in **both civil and military registers**.

The fleet of In-Service aircraft is increasing and therefore the number of incidents and repairs related with Fatigue will continue growing in the coming years. The Fatigue &DT justifications are key to maintain the airworthiness of the fleet.

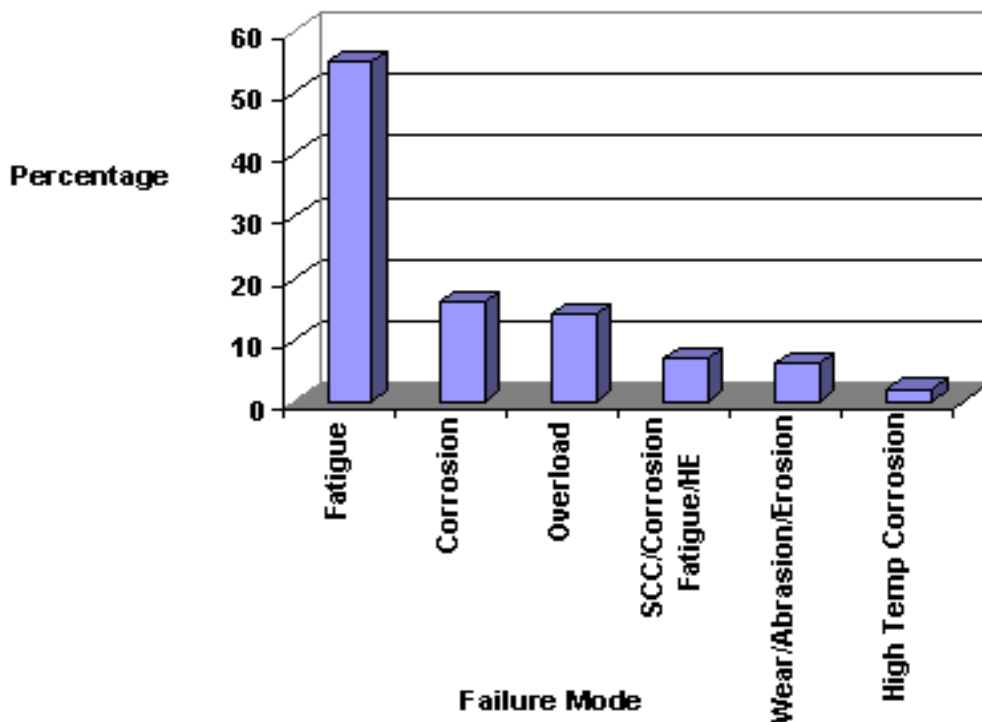
History of structural Fatigue

The Fatigue and Damage Tolerance (F&DT) task:

The Fatigue specialist function starts with the preliminary design and ends when the aircraft of the fleet are retired from service (“Cradle to grave”). The significant number of activities related with the F&DT discipline means that it is embedded in a **multi-disciplinary environment** that requires interaction with many other areas of expertise affecting the **structural integrity, safety and economic aspects** of the operation. Another relevant aspect of the discipline is related with the definition of the structural maintenance tasks.

History of structural Fatigue

Aircraft Component Failures



“Fatigue is an insidious process which searches out small design imperfections which are not apparent from the most detailed calculations. For this reason, clearance by calculation is normally acceptable only for simple components. Fatigue lives which are not underpinned by component tests must include an allowance for the added uncertainty.”

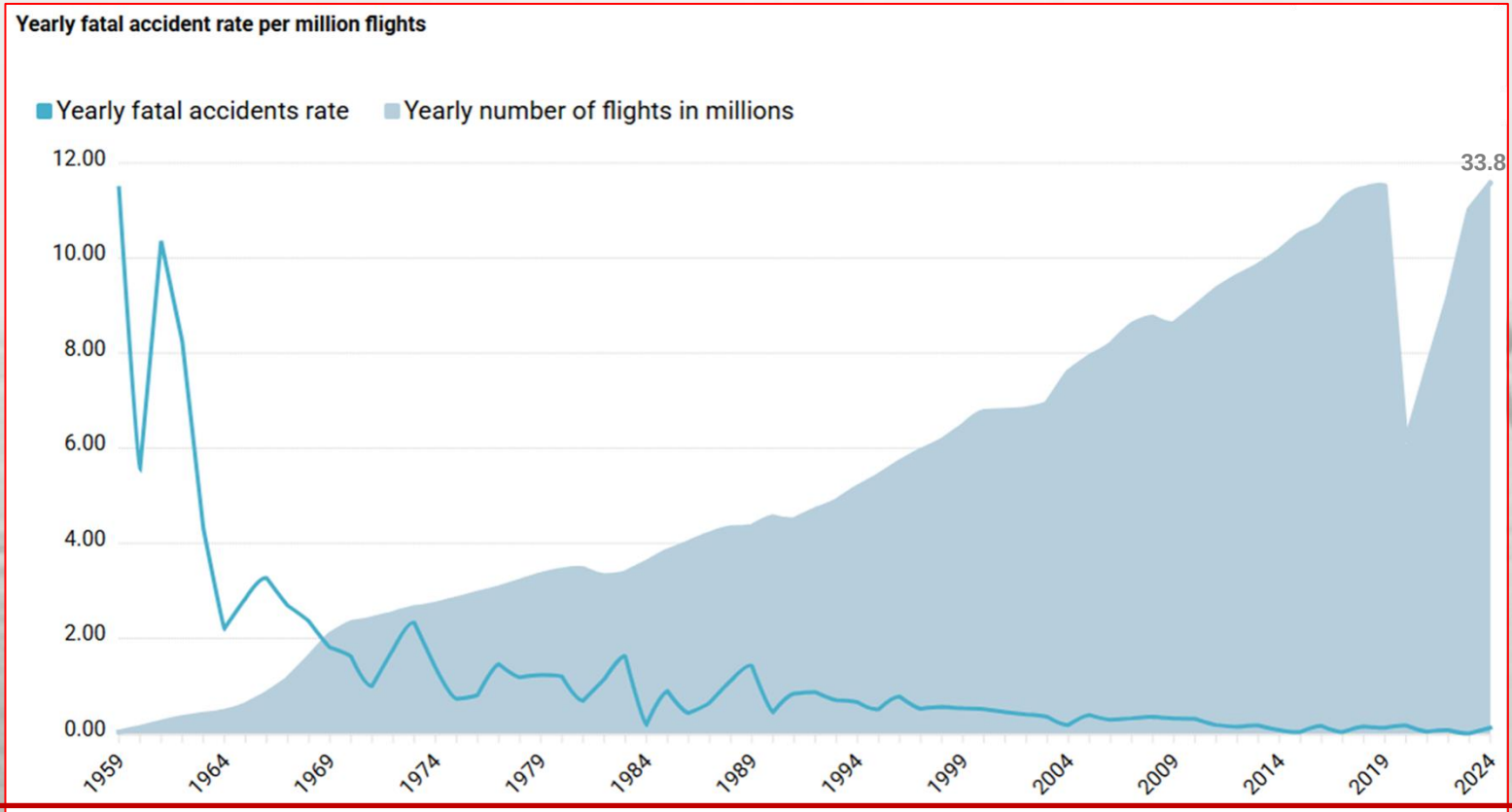
DEF STAN 00-970

LEAFLET 35 Part 1 Section 3

FATIGUE SAFE-LIFE SUBSTANTIATION

Structural Fatigue can be defined as the progressive deterioration of the strength of the materials due to in-service cyclic loads.

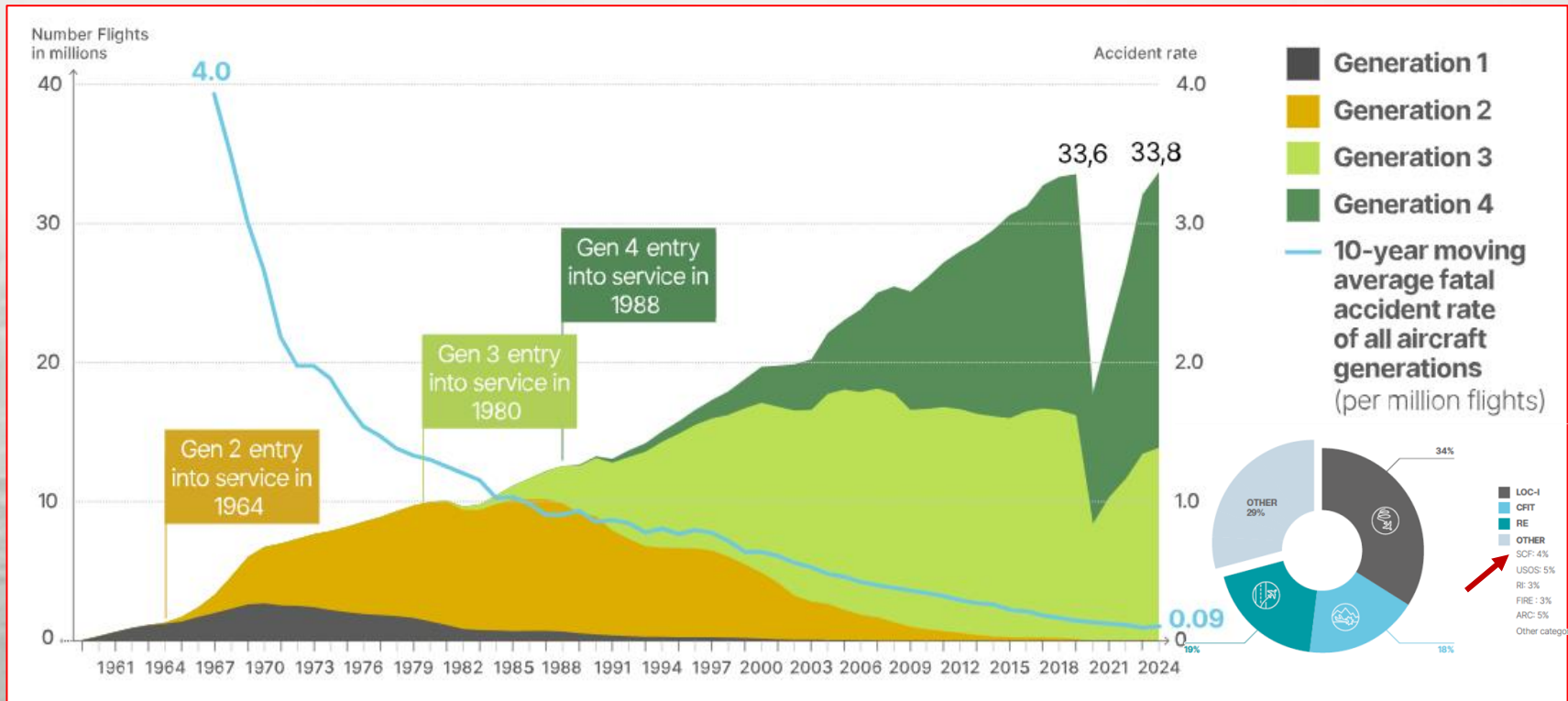
History of structural Fatigue



Reduction of accident rates is due to:

- an effective regulation,
- a strong safety culture,
- improvements in training,
- technological advances, particularly aircraft systems

History of structural Fatigue

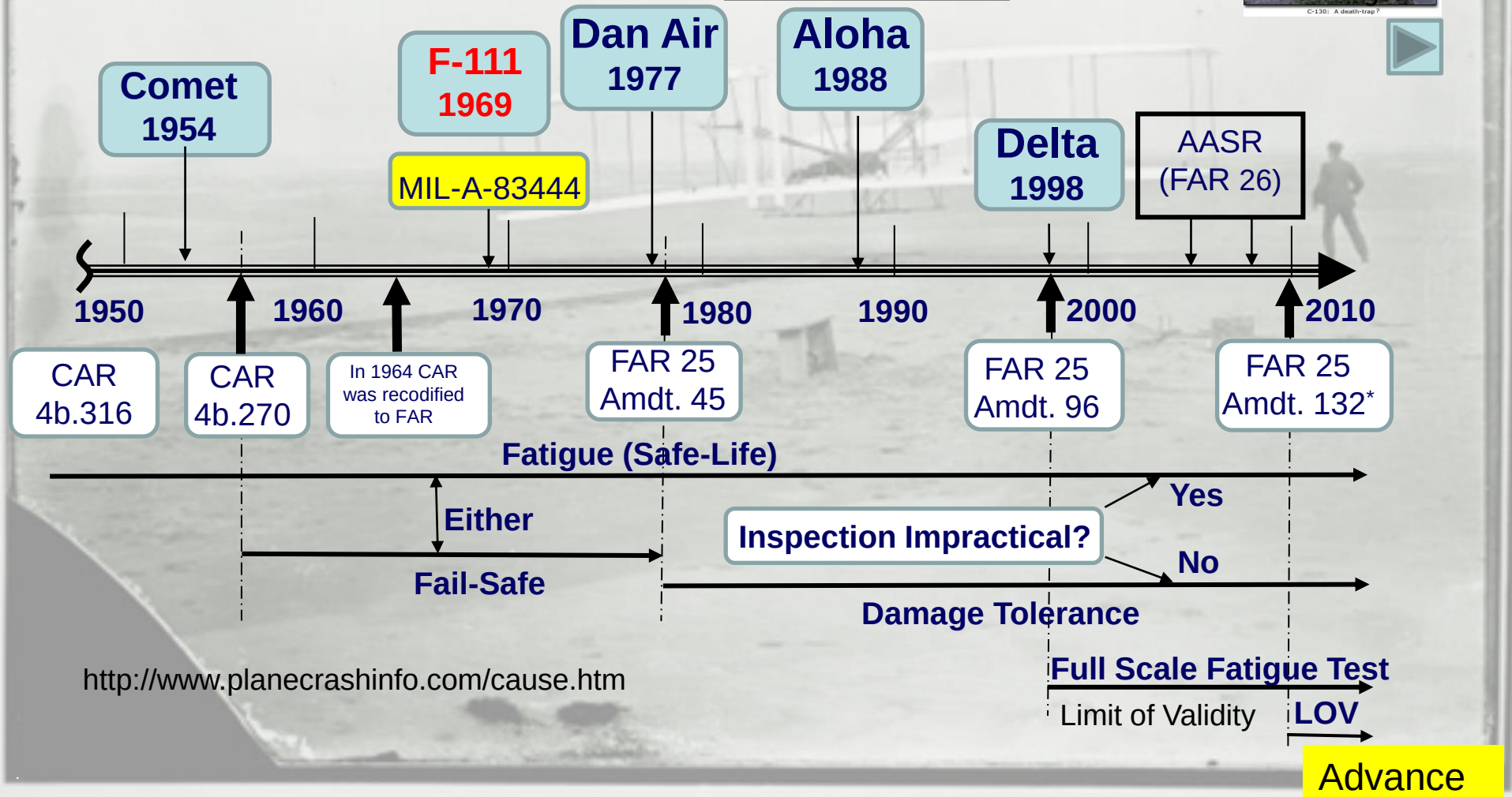


The aircraft designed taking into account the Fatigue and Damage Tolerance requirements have improved dramatically the safety records.

LOC-I
CFIT
RE
SCF

Loss of Control in Flight
Controlled Flight Into Terrain
Runway Excursion
System/Component Failure

History of structural Fatigue. Evolution of Certification Requirements after some events.

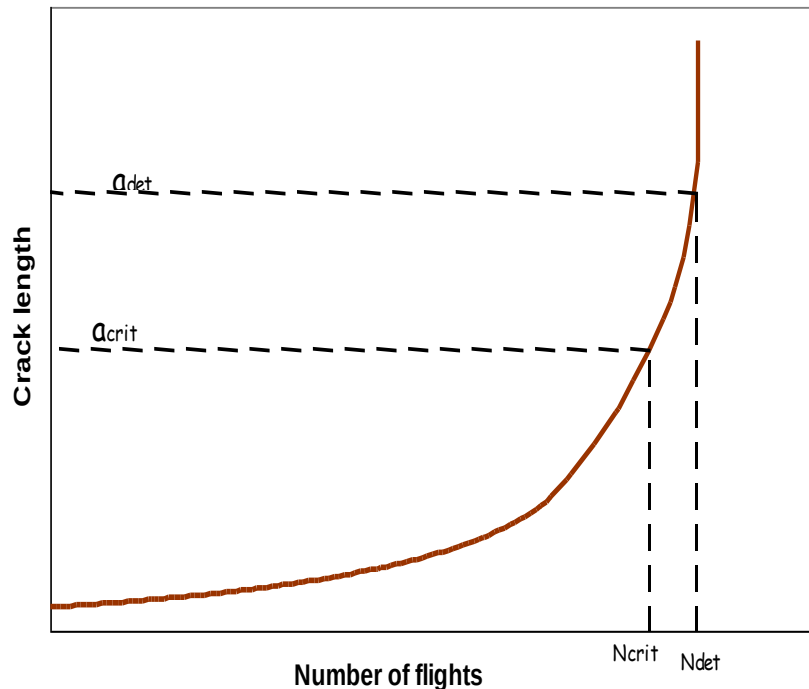


<http://www.planecrashinfo.com/cause.htm>

History of structural Fatigue

TYPES OF STRUCTURE (Some examples)

"SAFE-LIFE" STRUCTURES



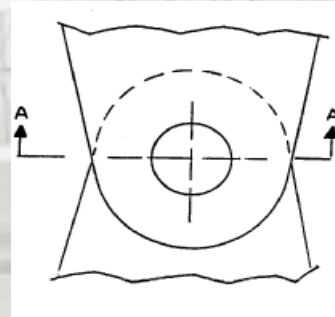
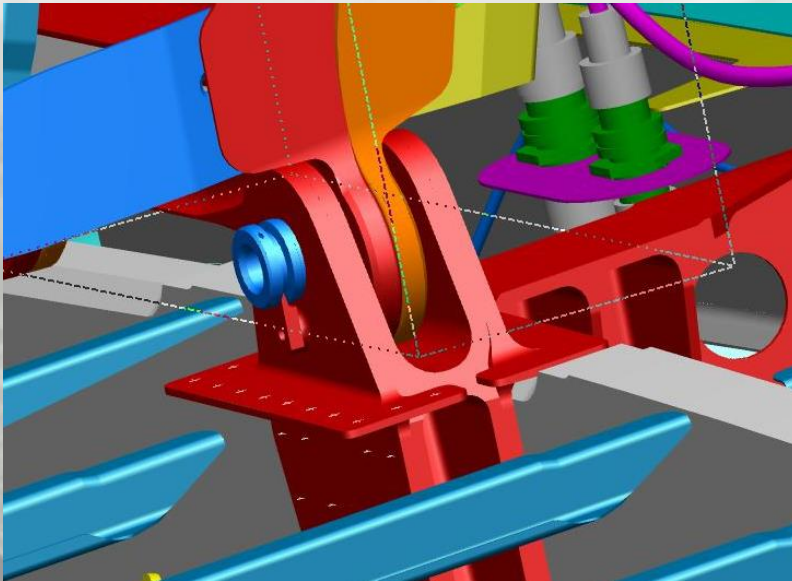
- LIMITED LIFE TIME. ITEMS ARE DISCARDED AFTER REACHING THE LIFE GOAL.
- NOT ENOUGH RESIDUAL STRENGTH CAPABILITY WHEN DAMAGED.
- INSPECTION IS NOT FEASIBLE.
- CIVIL AUTHORITIES ONLY ACCEPT LANDING GEARS IN THIS CATEGORY, BUT IT IS FREQUENT IN COMBAT AIRCRAFT DESIGN.

Safety by retirement of components

back

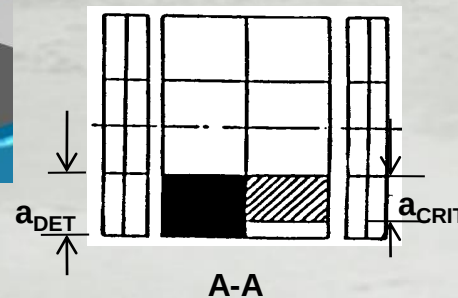
History of structural Fatigue

TYPES OF STRUCTURE (Some examples)

"FAIL-SAFE" STRUCTURES

➤ ENOUGH RESIDUAL STRENGTH CAPABILITY WHEN ONE OF THE LOAD PATHS IS BROKEN

➤ INSPECTIONS ARE NOT SPECIFIED.



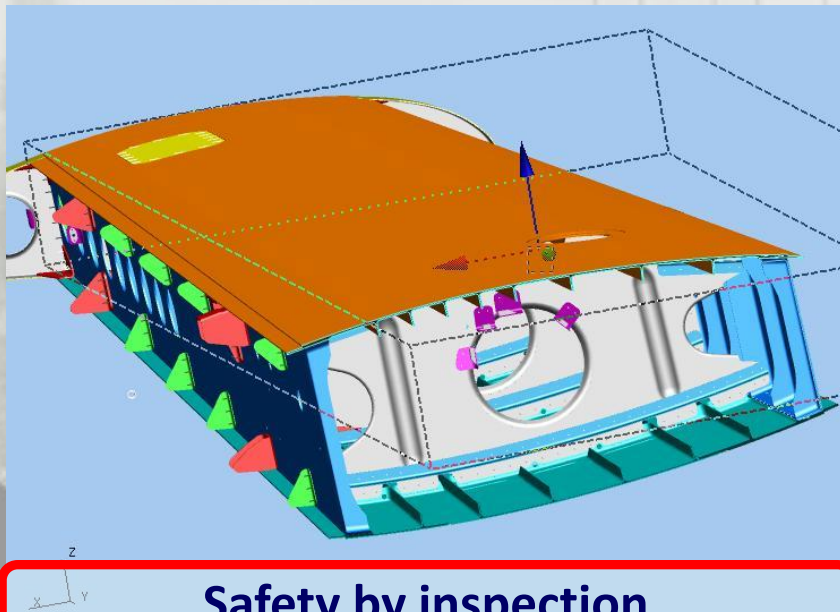
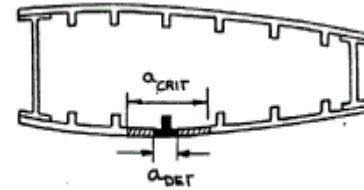
Concept later substituted by the Damage Tolerance Approach.

Safety by design (?)

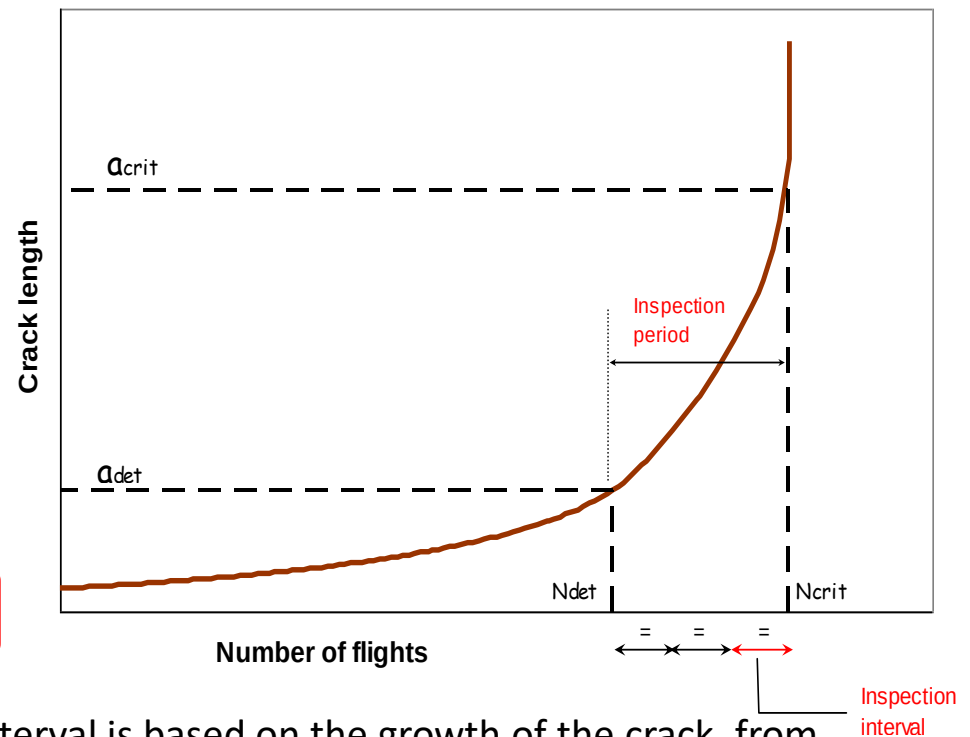
History of structural Fatigue

TYPES OF STRUCTURE (Some examples)

SINGLE LOAD PATH DAMAGE-TOLERANT



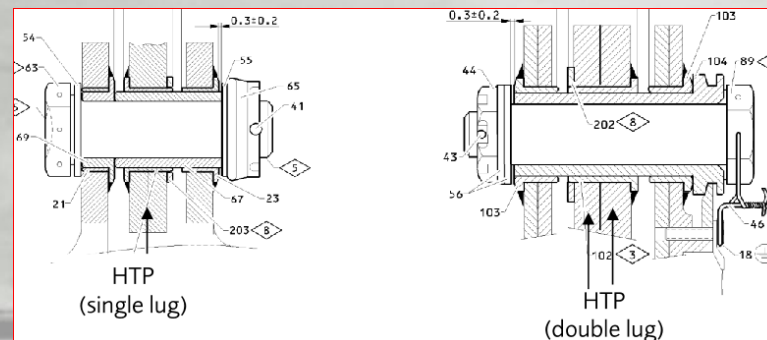
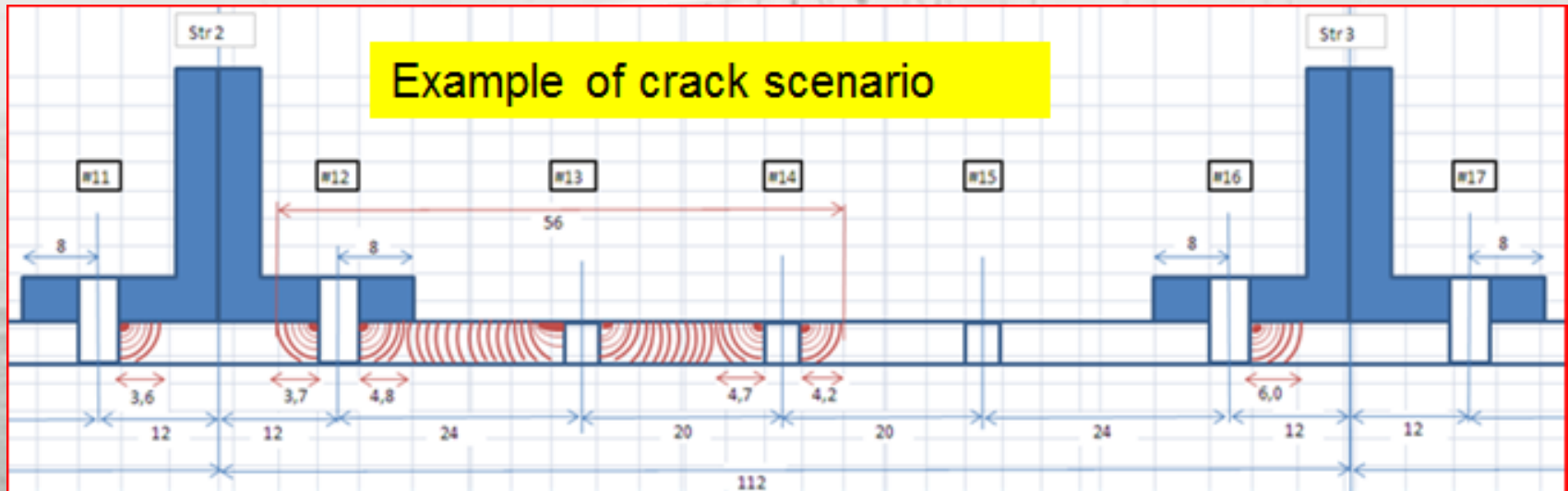
Safety by inspection



For the Single load path design the inspection interval is based on the growth of the crack, from its detectable size up to its critical size.

History of structural Fatigue

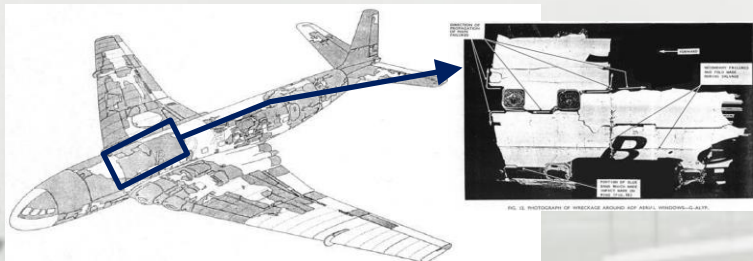
Current Civil regulations promote Damage Tolerant Multiple Load path designs against Single Load paths and Safe life designs (Except Landing Gears). The crack growth and residual strength analyses supported by tests are the accepted means of compliance.



History of structural Fatigue

https://www.youtube.com/watch?v=_BZnn5OYcBc

Comet. G-ALYP Wreckage recovered from the sea (1954)



The quasi-square windows had inadequate radii:
Detail design is key in fatigue. Stress concentrations to be avoided in highly stressed areas

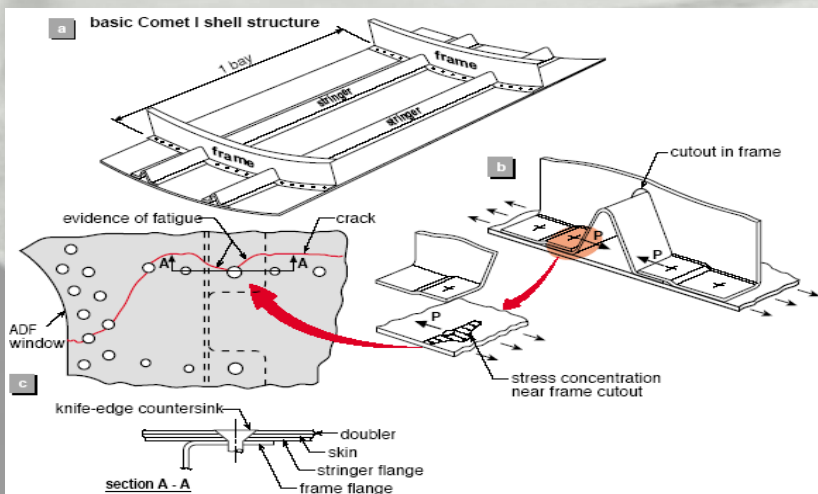


Fig. 5 Details of the probable failure origin of service aircraft Comet G-ALYP (Swift, 1987)

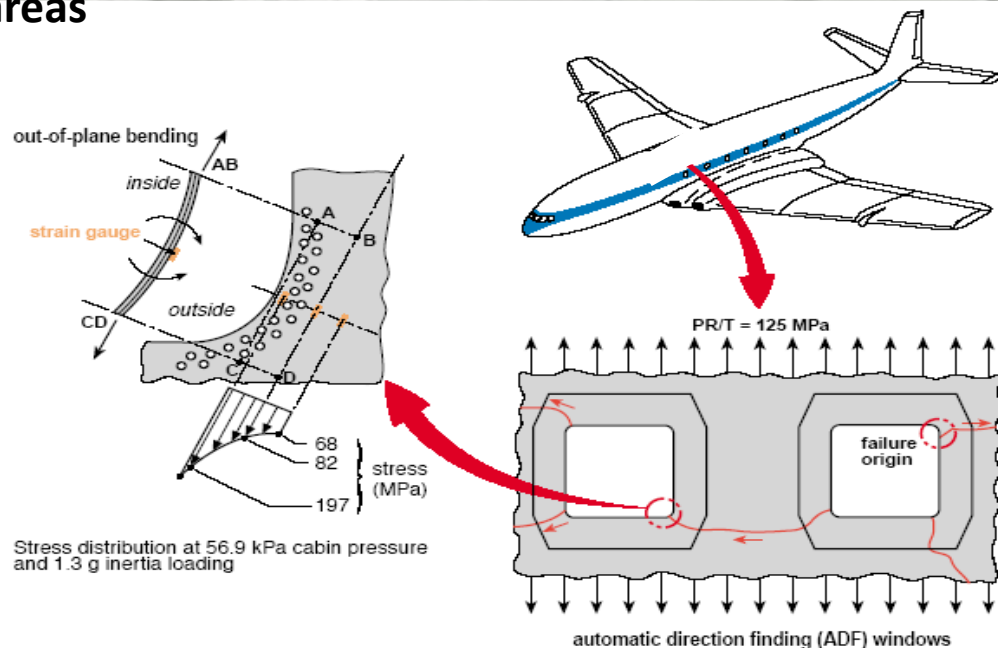
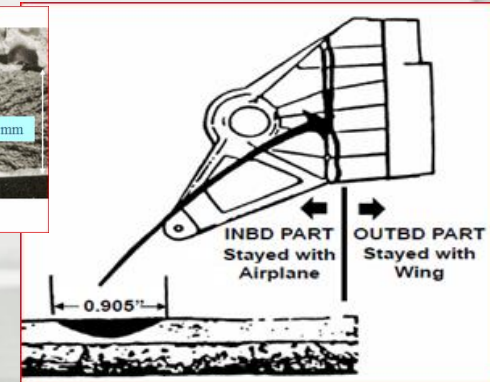
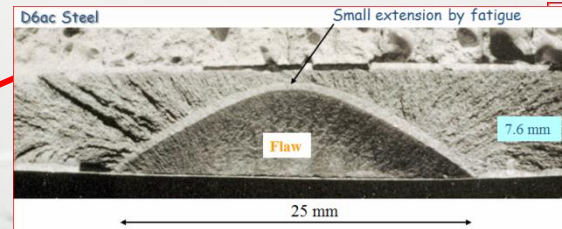
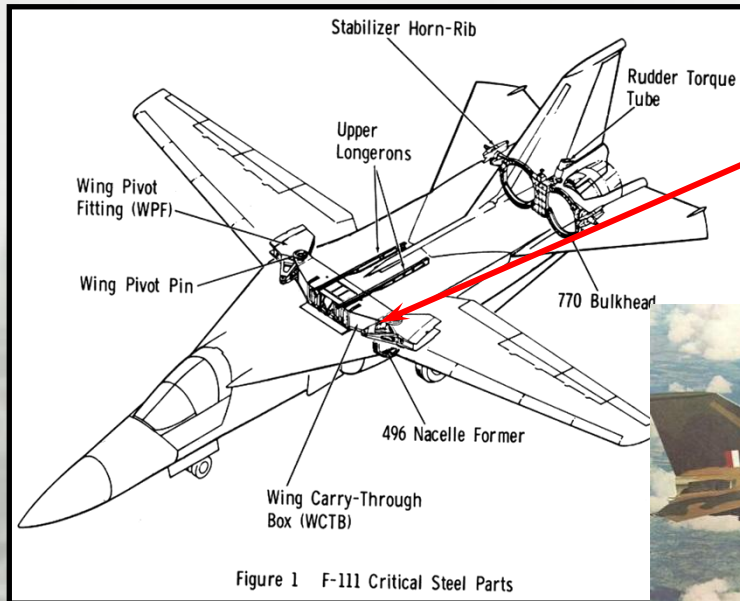


Fig. 4 Probable failure origin of service aircraft Comet G-ALYP: stress distribution obtained from repaired test aircraft, Comet G-ALYU (Swift, 1987)

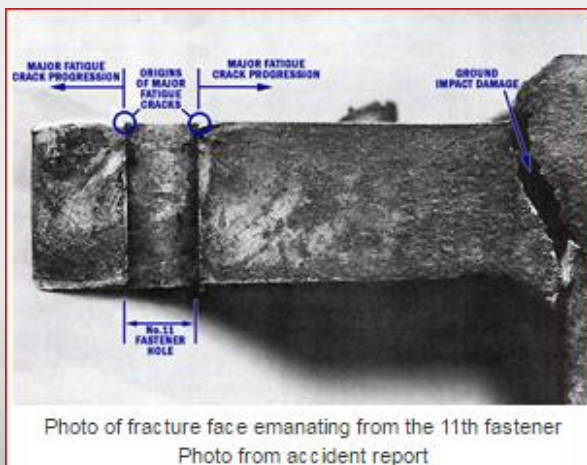
History of structural Fatigue

F-111 crashed due to manufacturing defect (1969)

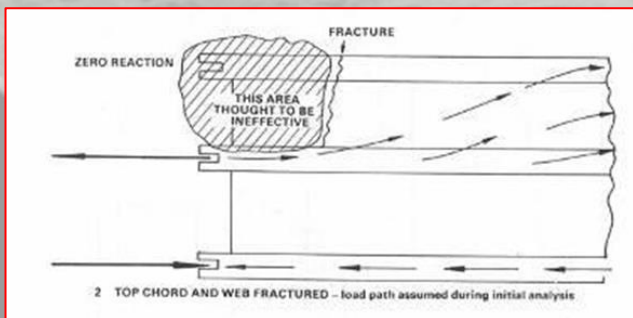
The undetected manufacturing defect propagated until critical size in a 4.0g maneuver... after accumulating only 100 flight hours!

Modification of MIL-A-83444 "Airplane Damage Tolerance Requirements": Accidental damage effects cannot be ignored

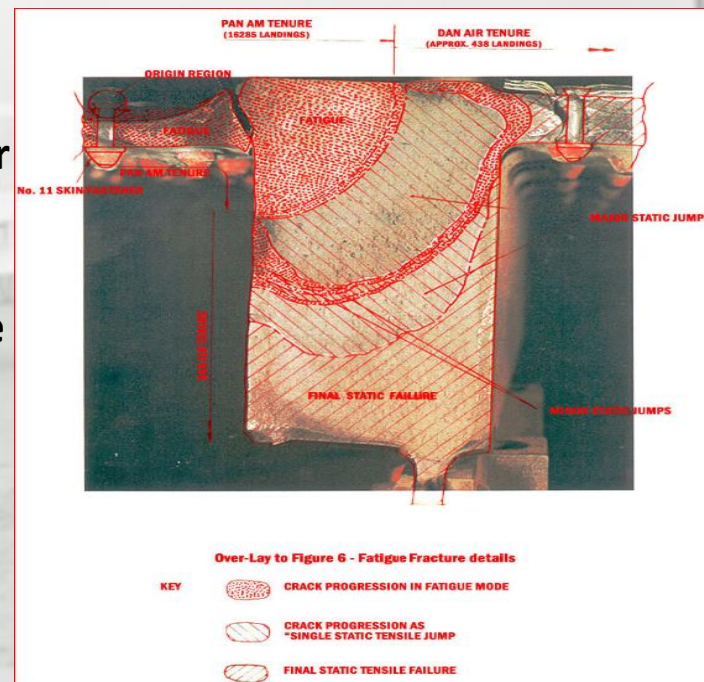
History of structural Fatigue



The fail-safe HTP rear spar upper chord failed prior to the accident due to long-term fatigue damage (a fatigue crack had existed for about 7,200 flights).



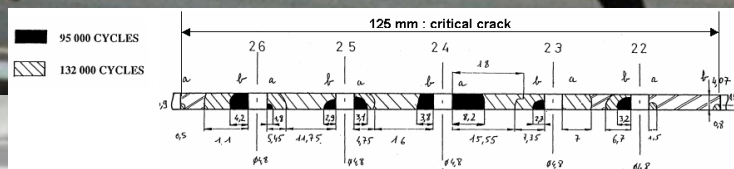
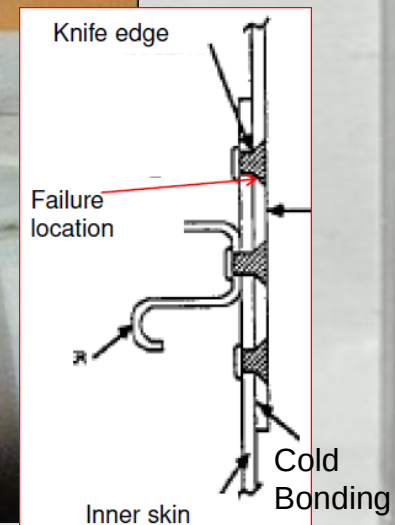
B707 DAN AIR 1977 (Lusaka)



The remaining structure was not able to sustain the flight loads for long enough for the failure to be detected under the normal maintenance. The inspection program was not adequate to detect partial cracks in the top chord of the rear spar.

Probable cause: "Failure of the Aloha Airlines maintenance program to detect the presence of significant disbonding and fatigue damage which ultimately led to failure of the riveted lap joint at S-10L and the separation of the fuselage upper lobe."

737 ALOHA accident 1988 – WIDESPREAD FATIGUE DAMAGE



**History of structural Fatigue: Some recent incidents.
Importance of the in-service experience!**



History of structural Fatigue

Fatigue in composites?

All in-service events shown previously are for metallic structure, but what about composite?

No significant fatigue in-service events reported in composite structure after more than 30 years of in-service experience

Composites are not so sensitive to fatigue for the typical in-plane strains; however the out of plane loads have a significant impact!

The current design follows a no-initiation and no growth concept by limiting the strain levels.

For composites it is necessary to take proper design precautions to ensure that fatigue is not an issue!

The fibres failure under tension loads and peeling or delamination damages under compression loads and the linear response plus the lack of local plasticity is the main difference of the anisotropic composites. **Natural damages are initiated in composites due to the out-of-plane stresses originated by local geometric effects:** Stringers run-outs, plies drop-offs, unfolding of radius or eccentricities.

History of structural Fatigue

Fatigue in composites?

Consideration of accidental, manufacturing, assembly and in-service damages:

Composites are susceptible to experience damages during manufacturing, assembly processes, transport and in-service operation, such as voids, porosity, delaminations, impacts, lightning strike, etc. leading to in-plane stress concentrations affecting the static strength. Some of these damages can be detected (and repaired) by the inspections of every single manufactured part before Entry Into Service (EIS) but some others not because are smaller than:

- BVID (**B**arely **V**isible **I**mpact **D**amage) for low energy accidental impacts
- Minimum flaws allowed by the quality control inspections for other manufacturing damages.

So, the **fatigue demonstration in composites is twofold:**

- No-initiation of natural damages during the service life (similar to metals)
- No-growth of non-detectable damages during the service life (specific of composites).

In addition, and unlike metals, **Ultimate Load capability after fatigue of BVID and other non-detectable damages needs to be demonstrated in composites.**

Structural Fatigue Introduction

Introduction to some of the Fatigue most usual terms

‘Fatigue critical structure (FCS)’ is structure that is susceptible to fatigue cracking that could lead to a catastrophic failure of an aircraft.

‘Safe-life’ of a structure is that number of events such as flights, landings, or flight hours, during which there is a low probability that the strength will degrade below its design ultimate value due to fatigue cracking.

‘Design service goal (DSG)’ is the period of time (in flight cycles or flight hours, or both) established at design and/or certification during which the aircraft structure is reasonably free from significant cracking.

‘Principal structure element (PSE)’ is an element that contributes significantly to the carrying of flight, ground, or pressurisation loads, and whose integrity is essential in maintaining the overall structural integrity of the aeroplane.

Structural Fatigue Introduction

Introduction to the Fatigue most usual terms

PRINCIPAL STRUCTURE ELEMENT (PSE).

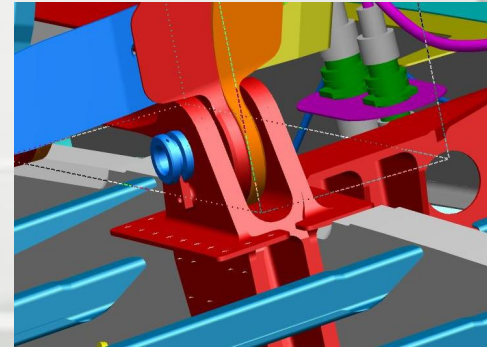
- PART OF THE PRIMARY STRUCTURE.
- IT CAN PRODUCE A CATASTROPHIC FAILURE.
- IT MUST BE DAMAGE TOLERANT, OR “SAFE-LIFE”.

SIGNIFICANT STRUCTURE ELEMENT (SSE).

- PART OF THE PRIMARY OR SECONDARY STRUCTURE.
- IT CAN DIRECTLY OR INDIRECTLY PRODUCE CATASTROPHIC FAILURE.
- IT MUST BE DAMAGE TOLERANT OR “SAFE-LIFE”.

SIGNIFICANT STRUCTURE ITEM (SSI).

- COMMON ACCESS FOR ALL ITS ELEMENTS.
- A PSE IS PART OF A SSI, BUT NOT THE OTHER WAY AROUND



Structural Fatigue Introduction

Introduction to the Fatigue most usual terms

PRIMARY STRUCTURE

- IT CARRIES FLIGHT, GROUND OR PRESSURIZATION LOADS



SECONDARY STRUCTURE

- IT CARRIES AERODYNAMIC OR INERTIA LOADS GENERATED ON OR WITHIN THE STRUCTURE



Structural Fatigue Introduction

Introduction . Civil Standards and Regulations

Airworthiness system in USA

Code of Federal Regulations (CFR)

Title 14: Aeronautics and Space

Chapter I : Federal Aviation Administration (FAA)

Subchapter C: Aircrafts – Parts 21-59

<https://www.faa.gov/>

FAR 25.571: Damage tolerance and fatigue evaluation of structure

Reminder

Part 25: Airworthiness standards: Transport category airplanes

Part 26: Continued airworthiness and safety improvements for transport category airplanes

https://www.faa.gov/documentLibrary/media/Advisory_Circular/AC_25_571-1D_.pdf

Structural Fatigue Introduction

Introduction . Civil Standards and Regulations

Airworthiness system in Europe

Reminder

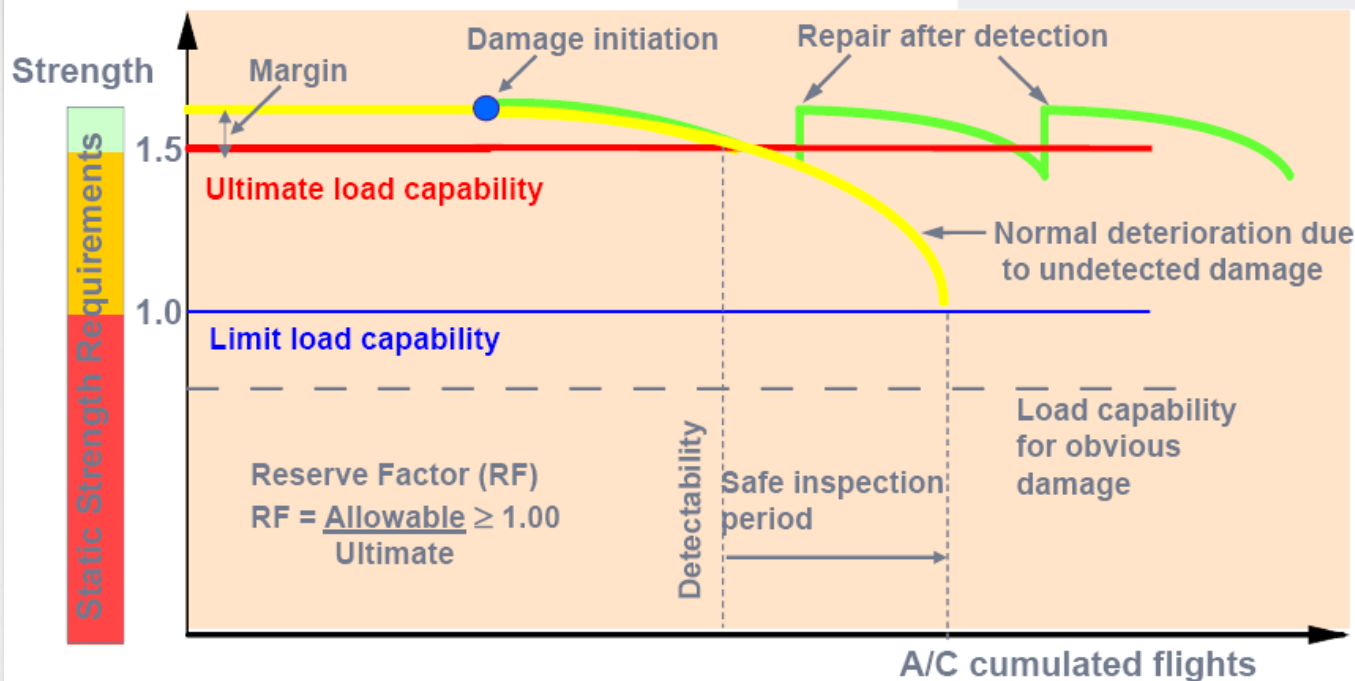
EASA is the European Authority similar to FAA.
Similar organization to FAR but in Europe the Parts are named Certification Standards (CS). Most CS have the same numbering than its related FAA part.

Most important CS related to fatigue: CS 25 , Paragraph 25.571

<http://www.easa.europa.eu/>

Structural Fatigue Introduction

Introduction . Civil Standards and Regulations



For Damage Tolerant designs Safety relies on inspections

Reminder

Fatigue & Damage Tolerance Requirements

The FAR/CS regulations, in Chapter 25.571, admit the fact that, due to non detected structural failures, the static strength can be lower than the ultimate load. In fact, such strength level decreases with the service life of the aircraft due to a progressive occurrence of the aforementioned damages. The requirement is that such strength level at least must correspond to the limit load (RESIDUAL STRENGTH).

Structural Fatigue Introduction

Introduction . Military Standards and Regulations

In the case of military aircraft, the design concept is different, and therefore the regulations for Fatigue and damage tolerance evaluation are also different.

For instance, some old regulations for military aircraft with Fatigue and damage tolerance requirements are:

- MIL-A-83444 “Airplane Damage Tolerance Requirements”. The initial specific regulation that contains the design requirements for damage tolerance applicable to the structure.
- MIL-A-87221 “Aircraft Structures. General Specification for”. It supersedes the previous one, it is more general, it contains and makes reference to all the rest.

The following regulations are more general and consider, in part, criteria of damage tolerance and durability:

- MIL-A-8866, “Airplane Strength and stiffness, Ground Tests”.
- MIL-A-8867, “Airplane Strength and stiffness, Reliability Requirements, Repeat Loads and Fatigue”.
- MIL-STD-1530, “Aircraft Structural Integrity Program Airplane Requirements”.
- JOINT SERVICE SPECIFICATION GUIDE (JSSG-2006). It supersedes MIL-A-87221.

Reminder

Structural Fatigue Introduction

Introduction . Military Standards and Regulations

Other regulations different to the MIL are for example:

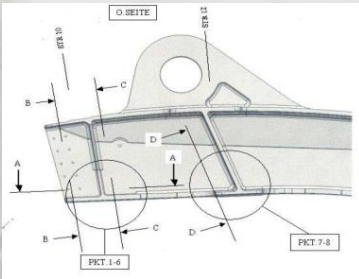
- AVP 970, used along with the MIL for the C-101.
- [DEF-STAN 00-970](http://www.dstan.mod.uk/data/ix00-970.htm) (<http://www.dstan.mod.uk/data/ix00-970.htm>)
- [AUSTRALIAN STANDARDS](http://www.defence.gov.au/dmo/asd/dapa_pub/Display.asp?GRP=AAP7)
(http://www.defence.gov.au/dmo/asd/dapa_pub/Display.asp?GRP=AAP7)
- STANAG 4671 UAV Systems Airworthiness Requirements (USAR), for North Atlantic Treaty Organization (NATO) Military UAV Systems

Reminder

Structural Fatigue Introduction

The Fatigue and Damage Tolerance of structures is affected by many probabilistic variables coming from several related competences

All the variables have to be considered concurrently NOT sequentially!



Design details



Manufacturing quality



F&DT Material properties



Loads and Fatigue Spectra Generation



Environmental conditions & Aging



Accidental damage



Inspections and Maintenance



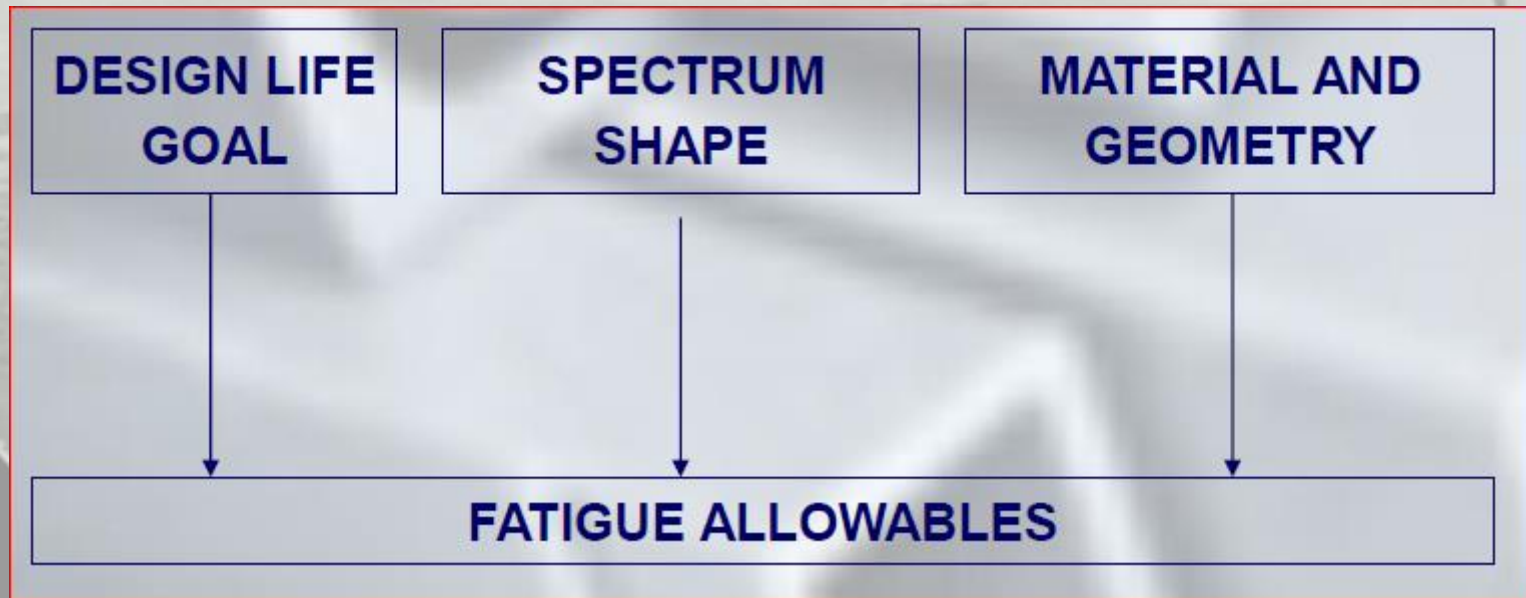
Probabilities / Scatter

Structural Fatigue Introduction

Fatigue endurance Design allowable values for the different materials.

Parameters affecting the fatigue.

The Fatigue allowable stress for an element can be defined as the maximum value that the tension limit stress can reach (on the point where it is maximum), so that the life goals are met.



Structural Fatigue Introduction

Fatigue endurance Design allowable values for the different materials.

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The following parameters have to be considered for each structural component:

Fatigue life goals in terms of Flight Hours (FH), Flight Cycles (FC) and Years (Y). The Fatigue life goals that are usually fixed on each of the design phases of the structure are the following:

- **Damage tolerant structure:** At least two times the Design Service Goal (DSG) or five times the Inspection Threshold target. (for Aluminum Alloys).
- **Safe-life structure:** five times the DSG. (for Aluminum Alloys).

Stress spectrum shape, and relationship between Fatigue and limit stresses

To determine the Fatigue allowables, a non-dimensional spectrum is used, in terms of load or stress related to the tensile limit load or stress of the element.

Materials and geometry

The Fatigue allowables depend also on the S-N curves of the material and the stress concentration factor K_t on the most loaded zone.

Structural Fatigue Introduction

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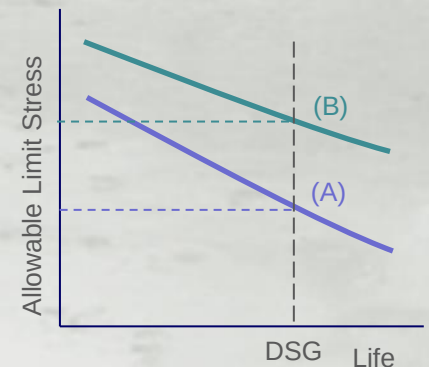
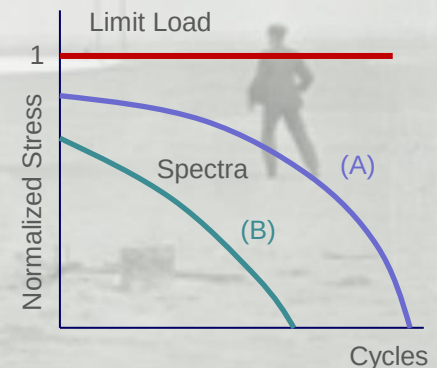
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Structural Fatigue Introduction

Design objectives. Safety and Economy. Inputs and deliverables in a multi-disciplinary environment.

Reparability, accessibility and inspectability objectives:

Repair or Rework of the Structure: In some cases, repair or rework of the structure can gain further life. The design has to allow for the repairs that are consequence of in service findings of fatigue cracks, accidental damages and corrosion. The level of structural optimization has to be according to an acceptable level of reparability. The original design has to consider details as the fastener pitch and edge distance with the allowances to permit fastener oversize without affecting significantly the F&DT lives. Also some precautions have to be taken to allow fastener installation in areas without them in case of riveted repairs are needed in the future.

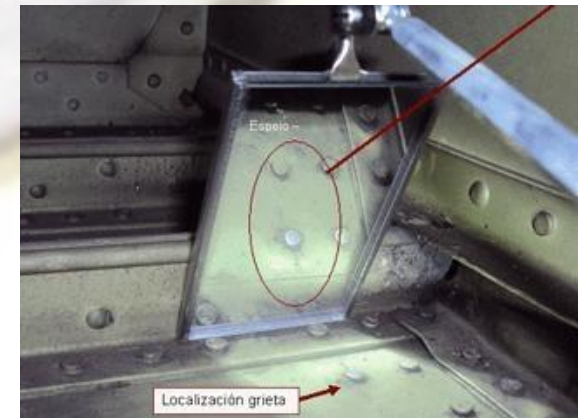
Structural Fatigue Introduction

Design objectives. Safety and Economy. Inputs and deliverables in a multi-disciplinary environment.

Reparability, accessibility and inspectability objectives:

Inspection: Detection of damage before it becomes dangerous is the ultimate control in ensuring the damage-tolerance characteristics of the structure so the access and good detectability of damages has to be considered in the design to allow acceptable, good compromise, inspection intervals.

Inaccessible Areas: Every reasonable effort should be made to ensure inspectability of all structural parts, and to qualify them under the damage-tolerance provisions. In difficult areas it should be demonstrated that crack propagation life is clearly sufficient for the design life.



Structural Fatigue Introduction

Types of damage

METALLIC STRUCTURE

- FATIGUE
 - IMPORTANT FOR SECURITY
 - MANUFACTURING DAMAGES
- ENVIRONMENTAL/Corrosion (*EAC)
 - IMPORTANT FOR THE ECONOMY
 - STRESS-CORROSION CASE
- ACCIDENTAL
 - IMPORTANT FOR THE ECONOMY
 - DISCREET SOURCE DAMAGES



Manufacturing quality

COMPOSITE MATERIALS STRUCTURE

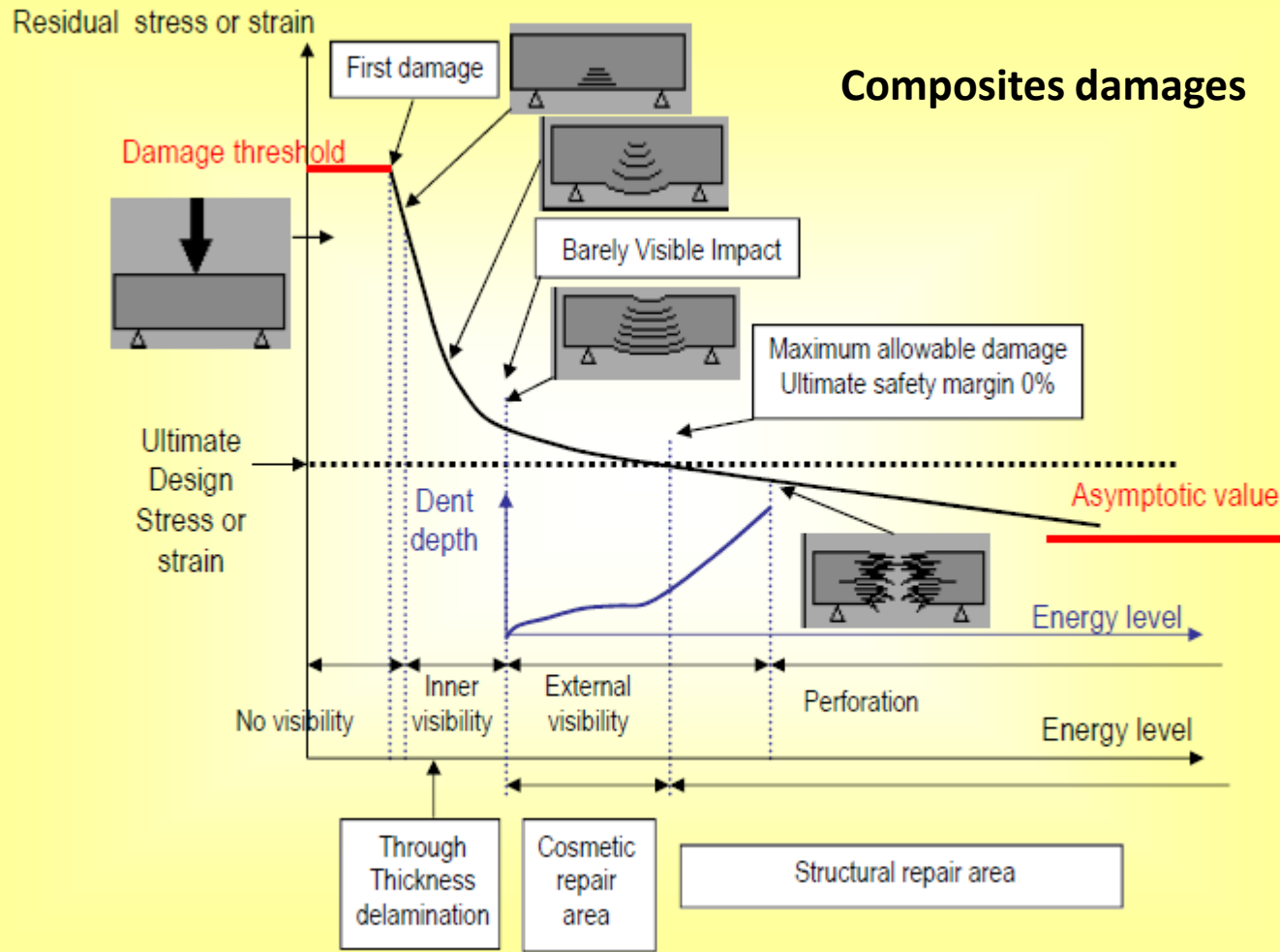
- FATIGUE
 - DESIGN CONCEPT OF NO INITIATION AND NO GROWTH
 - NOT IMPORTANT IN PRINCIPLE, EXCEPT FOR:
 - DESIGN DEFECTS
 - INAPPROPRIATE MANUFACTURING PROCESSES
 - PRODUCTION DEVIATIONS
- ENVIRONMENTAL/Corrosion
 - IMPORTANT FOR THE ECONOMY
 - HUMIDITY
 - AGING
- ACCIDENTAL
 - IMPORTANT FOR THE ECONOMY AND SECURITY
 - DAMAGES DURING THE MANUFACTURING PROCESS



Accidental damage

*EAC: Environmentally Assisted Cracking

Structural Fatigue Introduction



Valid for thickness $4 \text{ mm} < t < 10 \text{ mm}$

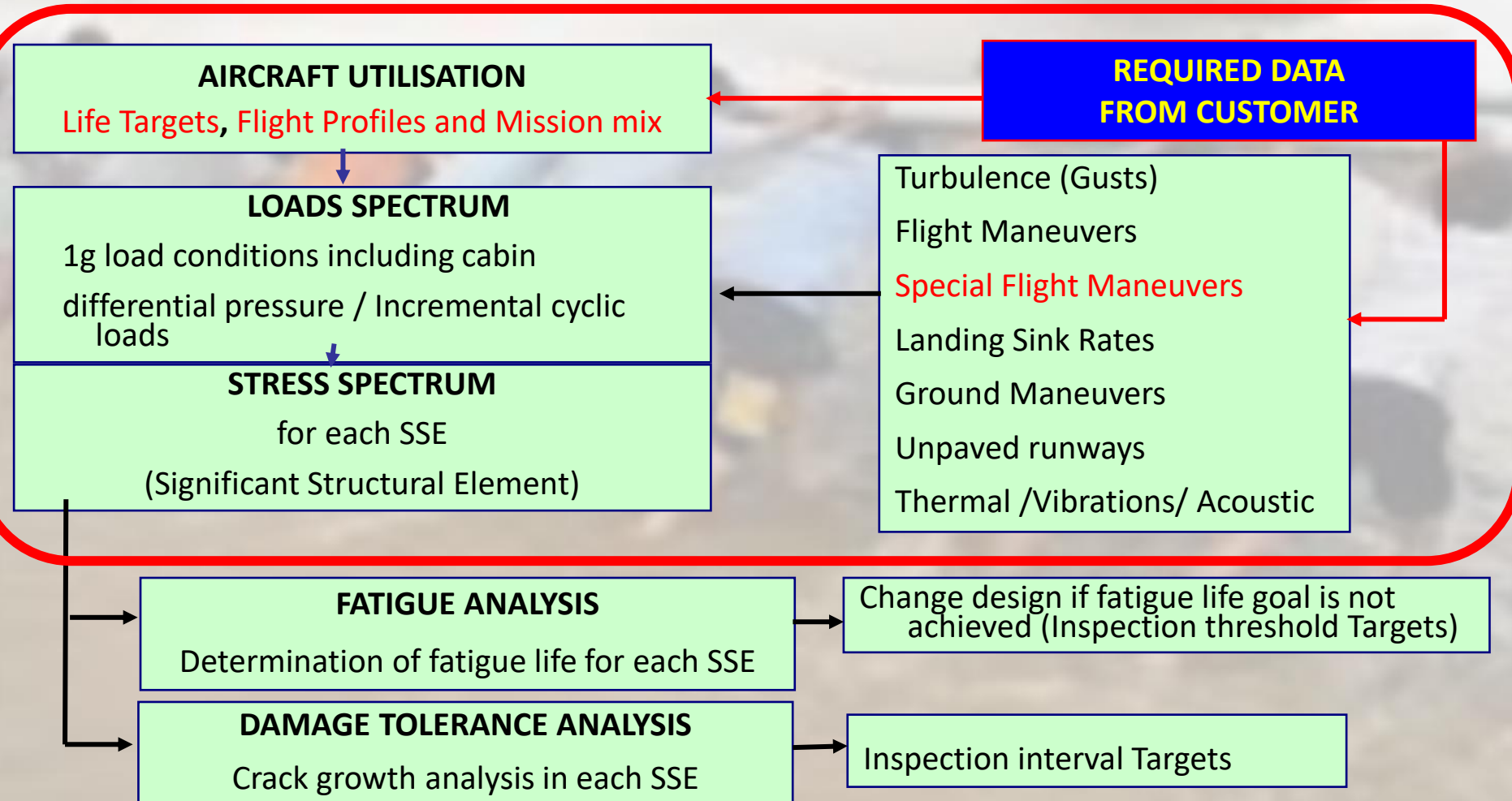
Types of damage

Composite Materials

Barely Visual Impact Damage (BVID): minimum detectable by scheduled inspection with a probability of detection of 90% with a confidence of 95%. It depends also on the inspection type.

Structural Fatigue Introduction

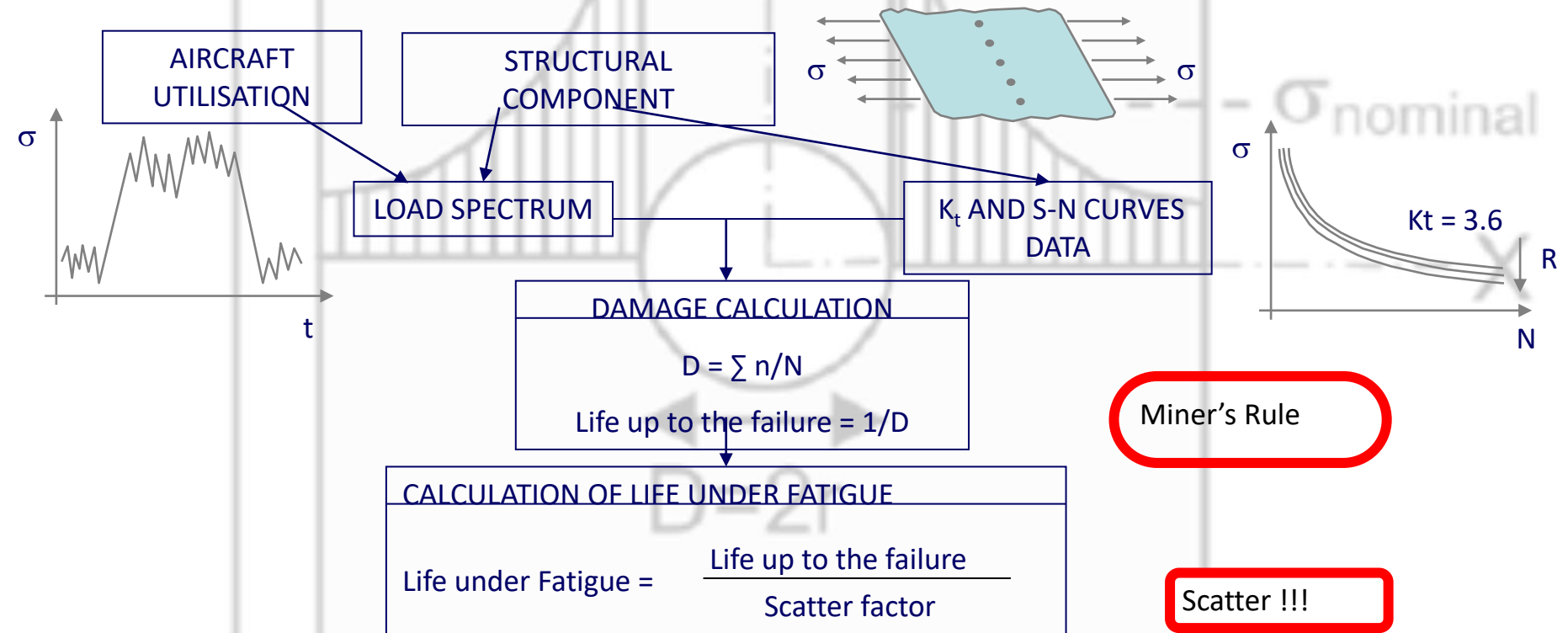
Aircraft Utilisation and Flight Profiles. Spectra determination



Structural Fatigue Introduction

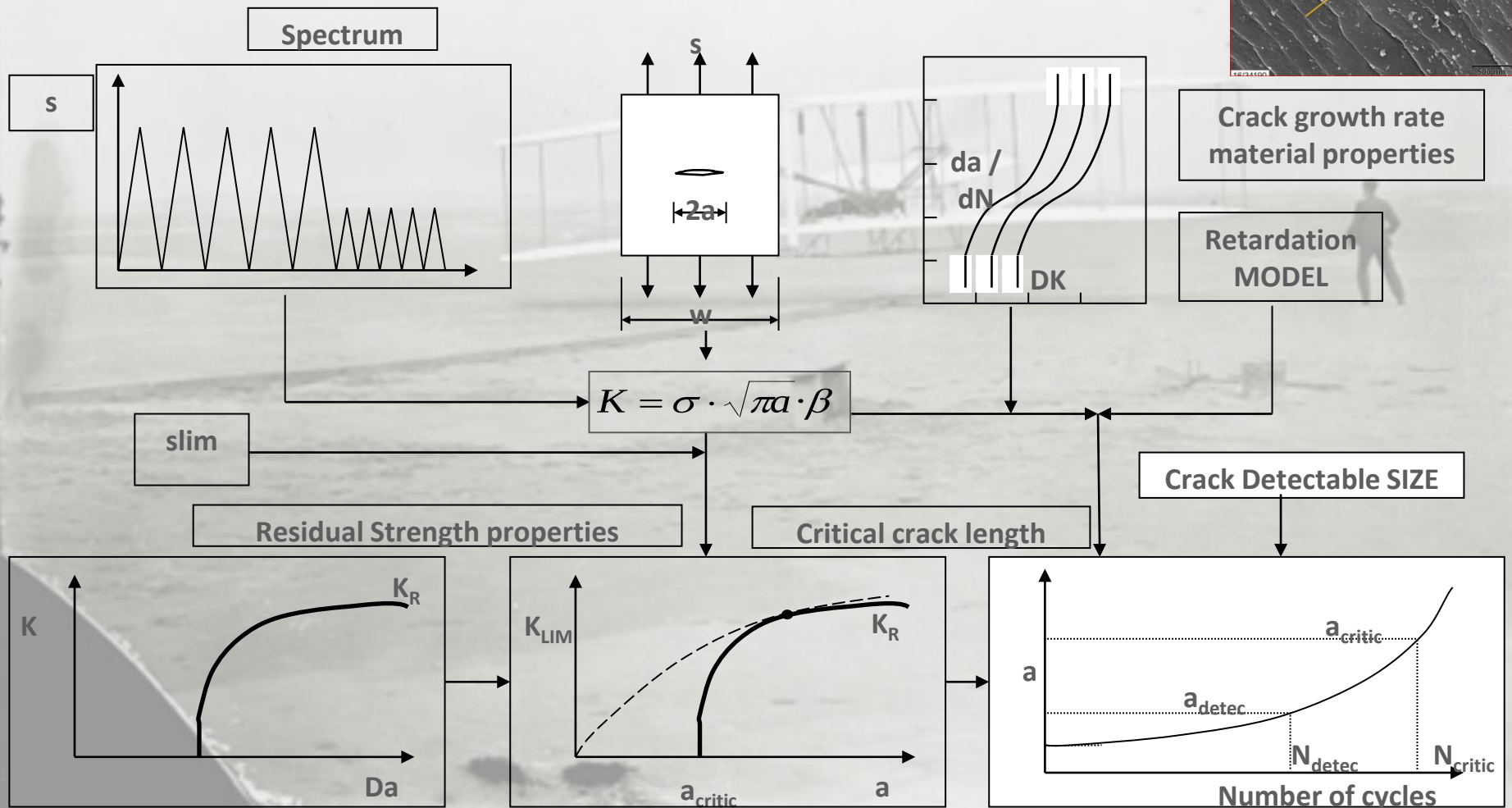
Basic Fatigue Concept: S-N Curves

SEQUENCE OF A FATIGUE ANALYSIS



Structural Fatigue Introduction

Damage tolerance analysis and supporting tests





*“We have forty million reasons for failure, but
not a single excuse”*

Rudyard Kipling

Thanks for your attention

Backup slides.

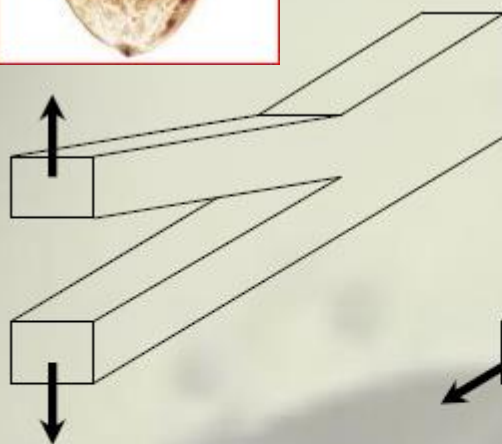
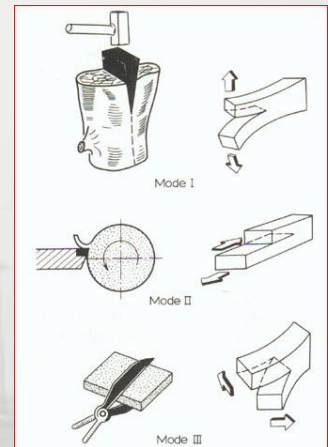
Fracture mechanics in the day to day life.

Also Bionic inspiration.

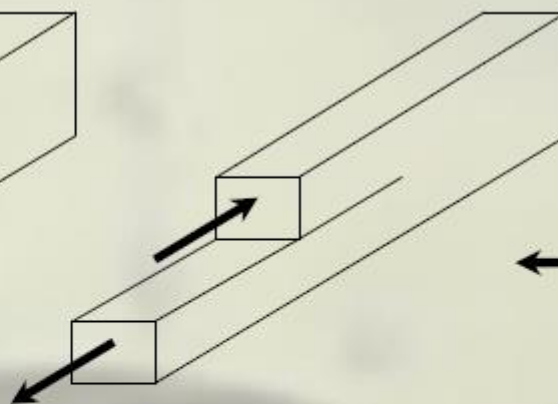


Backup slides.

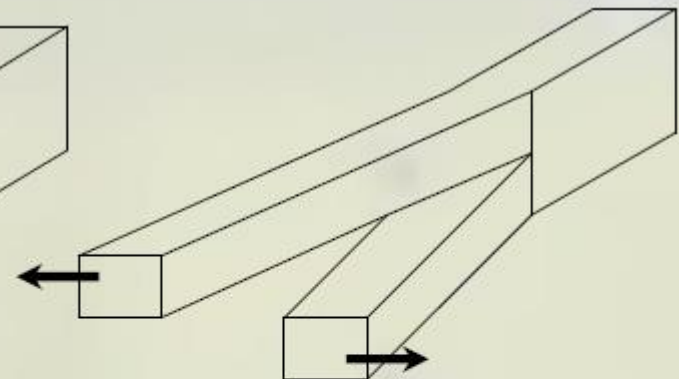
Fracture mechanics in the day to day life.



MODE I



MODE II



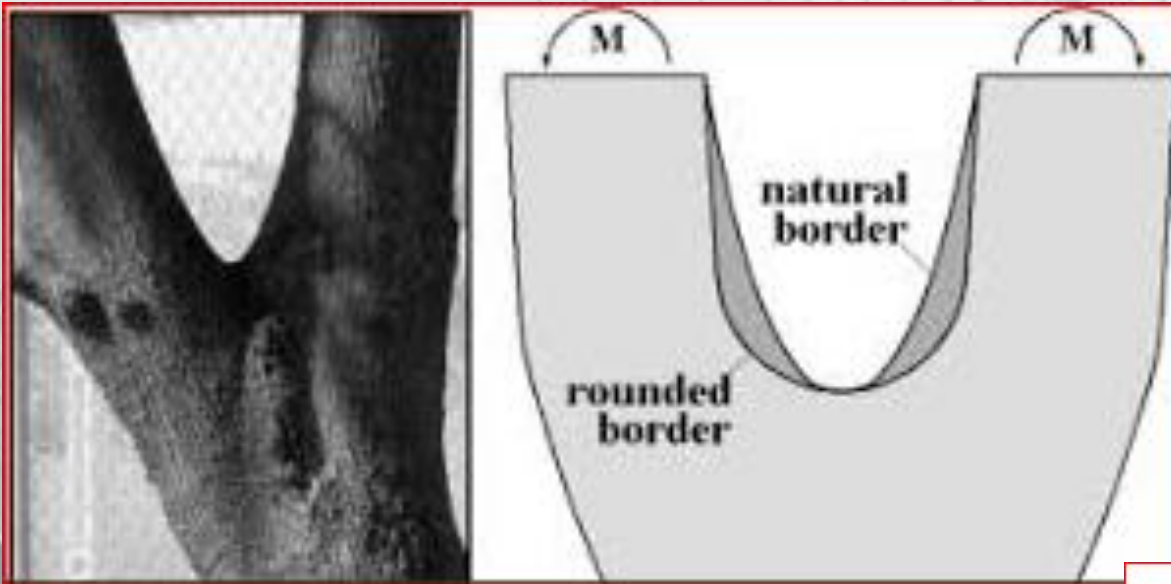
MODE III

CRACK PROPAGATION MODE

Backup slides.

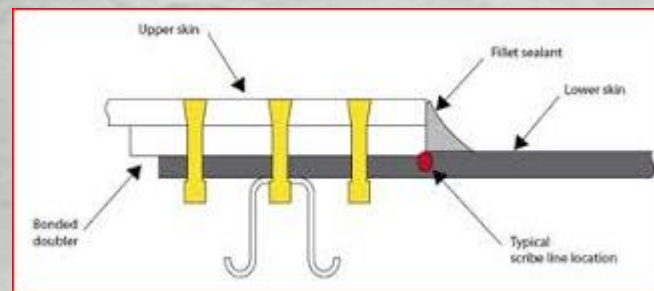
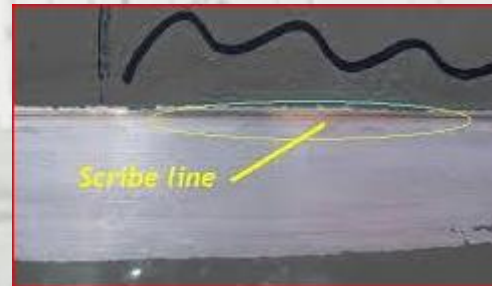
Fracture mechanics in the day to day life.

Bionic inspiration. Why Nature favors the big radiuses and soft transitions in many cases?



Backup slides.

Lesson learned: maintenance bad practices... Scribe Marks... or how to promote undesired cracks



Backup slides.

The cover is upside down in the right picture...

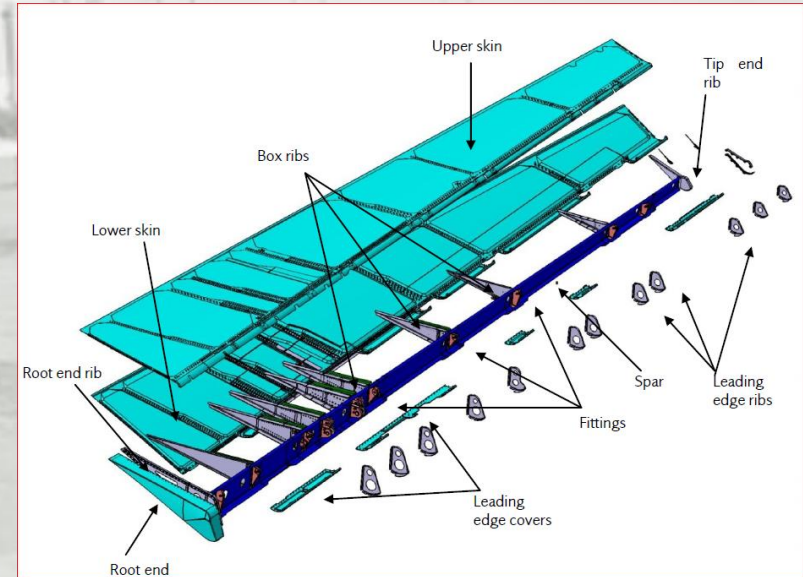
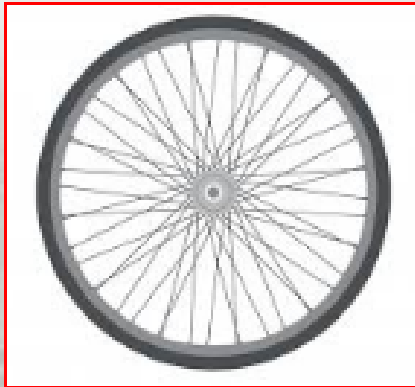
Would it be more resistant in static strength?

What about the fatigue durability?



Backup slides.

Can you say if the following elements are SLP, MLP, Safe Life, Damage Tolerant...?



Backup slides.

TIP: Find in Google images “fillet radius stress concentration factor” to understand the effect of the concentrating stresses.

